Problem Set 2

Applied Statistics / Quantitative Methods 1

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Question 1: Bribery and Driver Class

The data cross-tabulate driver class (upper, lower) against the outcome of a traffic encounter (not stopped, bribe requested, stopped/warning), for a total of $n = 42$ drivers.

1(a) Chi-square statistic

```r
##Make the contingency table
con_tab <- matrix(c(14, 6, 7,
                    7, 7, 1),
                  nrow = 2, byrow = TRUE,
                  dimnames = list(Class   = c("Upper", "Lower"),
                                  Outcome = c("Not stopped", "Bribe requested", "Stopped/warning")))
print(con_tab)


# Step 1: totals of each row, column and the overall test
row_tot <- rowSums(con_tab)
col_tot <- colSums(con_tab)
n       <- sum(con_tab)

print(row_tot)
print(col_tot)
print(n)
```

The observed counts, with row and column totals, are:

	Not stopped	Bribe requested	Stopped/warning	Total
Upper	14	6	7	27
Lower	7	7	1	15
Total	21	13	8	42

```r
# Step 2: expected counts = (row total * column total) / grand total
# multiplies every row total by every column total, giving the full grid at once
##this is the distribution according to expected counts under independence
expected <- (row_tot %*% t(col_tot)) / n
print(expected)
```

Under the null hypothesis of independence between class and outcome, the expected count in each cell is

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{n}.$$

This gives:

	Not stopped	Bribe requested	Stopped/warning
Upper	13.50	8.36	5.14
Lower	7.50	4.64	2.86

```r
# Step 3: chi-square statistic = sum of ((O)bserved - (E)xpected ^2 / E over all cells)
chi_sq <- sum((con_tab - expected)^2 / expected)
print(chi_sq)


#alternativly can also do this :
result <- chisq.test(con_tab)
print(result)
#extract the expected counts directly
result$expected
```

The chi-square statistic is

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \approx 3.79.$$

The built-in `chisq.test()` returns the same value, which confirms the manual calculation.

1(b) Significance at $\alpha = 0.10$

The table is 2×3, so the degrees of freedom are $(2-1)(3-1) = 2$. The p-value is

$$p = P(\chi^2_2 \geq 3.79) \approx 0.150.$$

At $\alpha = 0.10$ the critical value is $\chi^2_{2,\,0.10} \approx 4.605$. Since $3.79 < 4.605$ (equivalently, $p = 0.150 > 0.10$), we **fail to reject** the null hypothesis. This result means that there is not enough statistical evidence at the 10% significance level to suggest that the variables are related.

```r
# Degrees of freedom and p-value
df <- (nrow(con_tab) - 1) * (ncol(con_tab) - 1)
p_val  <- pchisq(chi_sq, df = df, lower.tail = FALSE)

print(df)
print(p_val)


#so the question asks what happens when the alpha is equal to 0.1 or 10%
alpha <- 0.1

if (p_val <= alpha){
  print("results are statistically significant. Reject the null hypothesis")
}else{
  print("results are not statistically significant. Fail to reject the null hypothesis")
}
```

1(c) Standardized residuals

For each cell the standard error is

$$\text{se}_{ij} = \sqrt{E_{ij}\left(1 - \frac{\text{row total}_i}{n}\right)\left(1 - \frac{\text{column total}_j}{n}\right)},$$

and the standardized residual is $(O_{ij} - E_{ij})/\text{se}_{ij}$. Each cell is shown below as observed (expected) [standardized residual]:

	Not stopped	Bribe requested	Stopped/warning
Upper	14 (13.50) [0.32]	6 (8.36) [−1.64]	7 (5.14) [1.52]
Lower	7 (7.50) [−0.32]	7 (4.64) [1.64]	1 (2.86) [−1.52]

```
#standard error for each cell
row_prop <- row_tot / n
col_prop <- col_tot / n
se <- sqrt(expected * ((1 - row_prop) %*% t(1 - col_prop)))
print(se)

#standardized residuals: (O - E) / se
st_re <- (con_tab - expected) / se
print(round(st_re, 3))
```

1(d) Interpretation

Under independence, the standardized residuals are approximately standard normal. At $\alpha = 0.10$ (two-tailed, 0.05 in each tail) the cutoff is ± 1.645.

- **Bribe requested:** the residuals come out at +1.64 for lower class drivers and -1.64 for upper class drivers. So just under the alpha (+/- 1.645)- meaning that lower class drivers get asked for bribes more than independence expected. However, still before the alpha cutoff so it is not statistically significant
- **Stopped/warning:** upper-class drivers were stopped or warned somewhat more than expected (+1.52), again below the cutoff.
- **Not stopped:** both residuals are small (± 0.32), so the two classes were not stopped at roughly the expected rates.

No cell reaches significance, which is consistent with the overall test in 1(b). The residuals add information that the overall test cannot: they show *which cells* are furthest from independence and in which direction.

Because the two classes have different sizes, the row percentages are easier to compare than raw counts:

	Not stopped	Bribe requested	Stopped/warning
Upper	51.9%	22.2%	25.9%
Lower	46.7%	46.7%	6.7%

Roughly the same share of each class was not stopped, but lower-class drivers were asked for bribes about twice as often (46.7% vs. 22.2%) and were rarely stopped or warned (6.7% vs. 25.9%). With only 42 observations, though, these differences are not statistically distinguishable from chance.

```
tab_out <- matrix(sprintf("%.0f (%.2f) [%.2f]", con_tab, expected, st_re),
                  nrow = nrow(con_tab), dimnames = dimnames(con_tab))
noquote(tab_out)

round(100 * prop.table(con_tab, margin = 1), 1)
```

Question 2: Women Leaders and Drinking-Water Facilities

2(a) Hypotheses

Do villages with a female-reserved leader get more new or repaired drinking-water facilities? The effect of reservation is the slope β_1 in

$$\text{water}_i = \beta_0 + \beta_1 \,\text{reserved}_i + \varepsilon_i.$$

- $H_0 : \beta_1 = 0$. Reserving the GP head position for women has no effect on the number of new or repaired drinking-water facilities.
- $H_1 : \beta_1 \neq 0$. Reserving the GP head position for women has an (unknown: either positive or negative) effect on the number of new or repaired drinking-water facilities.

2(b) Bivariate Regression of Hypotheses

Let x_i be `reserved`, y_i be `water`, and n the number of villages. The least-squares estimates are

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The residual variance uses $n - 2$ degrees of freedom because two parameters were estimated from the data:

$$\hat{\sigma}^2 = \frac{\sum_i \hat{\varepsilon}_i^2}{n-2}, \qquad \text{se}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum_i (x_i - \bar{x})^2}}.$$

The test statistic is $t = \hat{\beta}_1 / \text{se}(\hat{\beta}_1)$, compared with the critical value $t_{n-2,\,0.975}$, and the 95% confidence interval is $\hat{\beta}_1 \pm t_{n-2,\,0.975}\,\text{se}(\hat{\beta}_1)$.

```
women <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv")

#n = observations, x = reserved, y = water
x <- women$reserved
y <- women$water
nn <- length(y)

print(x)
print(y)
print(nn)

#Compute the means x_bar and y_bar.
```

```
x_bar <- mean(x)
y_bar <- mean(y)

print(x_bar)
print(y_bar)

#Intercept: b0_hat = y_bar - b1_hat*x_bar
#Slope: b1_hat = sum(x_i - x_bar)(y_i - y_bar) / sum(x_i - x_bar)^2
Sxy <- sum((x - x_bar) * (y - y_bar))
Sxx <- sum((x - x_bar)^2)
b1  <- Sxy / Sxx
b0  <- y_bar - b1 * x_bar

print(Sxy)
print(Sxx)
print(b1)
print(b0)

fitted_values <- b0 + b1 * x
residuals <- y - fitted_values
sigma2 <- sum(residuals^2) / (nn - 2)
se_b1  <- sqrt(sigma2 / Sxx)

print(fitted_values)
print(residuals)
print(sigma2)
print(se_b1)
```

- **Fitted Values ($\hat{\beta}_0 + \hat{\beta}_1$ * x):** prediction for each village
- **Residuals:** actual minus predicted
- **Residuals Variance (sigma2):** measure of how noisy the data is and nn - 2 instead of nn because we estimated two things (b0 and b1) from the data, which uses up two "degrees of freedom."
- **Standard Error (se_b1):** is the standard deviation of the estimate across different samples-turns the noise into uncertainty range

```
#t-statistic, critical value, two-tailed p-value
t_stat <- b1 / se_b1
t_crit <- qt(0.975, df = nn - 2)
p_value  <- 2 * pt(-abs(t_stat), df = nn - 2)

print(t_stat)
print(t_crit)
print(p_value)

# 95\% confidence interval
ci <- c(b1 - t_crit * se_b1, b1 + t_crit * se_b1)
print(ci)

results <- c(n = nn, b0 = b0, b1 = b1, se = se_b1,
             t = t_stat, t_crit = t_crit, p = p_value)
print(results)
```

```r
#alternatively, can do this too
summary(lm(water ~ reserved, data = women))
```

The p-value answers: If the null were true and reservation had no effect —¿ how likely is it that I'd see an estimate this far from zero? If $p < 0.05$, that's unlikely, so you reject H_0. Estimated regression coefficient is large relative to its standard error, meaning the relationship is statistically significant If $p \geq 0.05$, you fail to reject H_0. You don't "accept" it, you just lack enough evidence against it.

```r
if (abs(t_stat) > t_crit) {
  print("Reject the null hypothesises (H0: b1 = 0), as the t-stat was larger than the critical
    value")
  }else{
  print("Fail to reject H0- not enough statisticsal evidence  against it")
}
```

Quantity	Value
Number of villages n	322
Intercept $\hat{\beta}_0$	14.73832
Slope $\hat{\beta}_1$	9.25
Standard error se($\hat{\beta}_1$)	3.947746
t-statistic	[t]
Critical value $t_{n-2,\,0.975}$	2.343723
Two-tailed p-value	0.0197
95% CI for β_1	1.485608 17.019238

Since $p = 0.0197 < 0.05$ (equivalently, $|t| > t_{\text{crit}}$), we **reject** H_0. The p-value answers the question: if reservation truly had no effect, how likely would we be to see an estimate this far from zero? Here that is unlikely. Had $p \geq 0.05$, we would only *fail to reject* H_0, which is not the same as accepting it. The manual calculation can be checked against `summary(lm(water ~ reserved, data = women))`.

2(c) Interpret the Coefficient Estimate for Reservation Policy

The bivariate regression compares two groups of villages: those with a reserved seat (reserved = 1) and those without (reserved = 0). The regression is a way of making that comparison and testing whether the gap is real or due to chance.

- **Intercept ($\hat{\beta}_0$):** the average number of new or repaired water facilities in unreserved villages.
- **Slope ($\hat{\beta}_1$):** the difference between the two group averages (reserved minus unreserved).
- **Standard error:** how much $\hat{\beta}_1$ would vary across different samples of villages.
- **t-statistic:** how many standard errors $\hat{\beta}_1$ is from zero. A magnitude above roughly 2 means the difference is too large to attribute to chance.

- **p-value:** if H_0 were true, the probability of an estimate at least this extreme. A small value (here $p < 0.05$) means a difference this large is unlikely to be chance.

The estimated coefficient on `reserved` is 9.25: on average, villages in GPs reserved for women leaders have about 9 more new or repaired drinking-water facilities than villages in unreserved GPs. The p-value of 0.0197 is below 0.05, so we reject the null hypothesis and conclude there is statistically significant evidence that the reservation policy affects the number of drinking-water facilities. Because reservation was randomly assigned, this estimate can be interpreted as the causal effect of reservation.